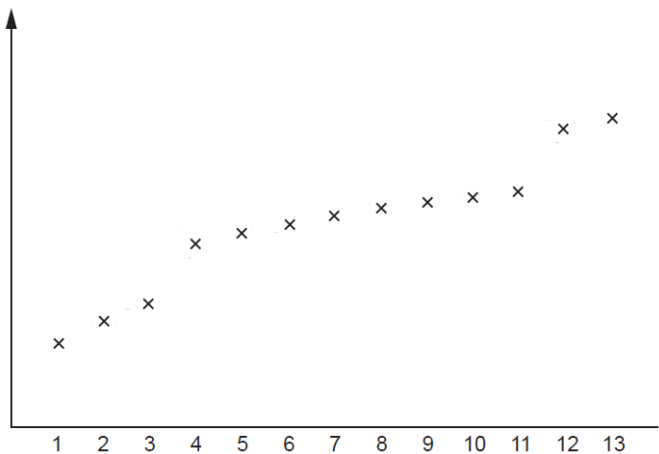


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)				1		1		
		(ii)	I	award (1) for any of following • general increase because there is a greater effective nuclear charge as more electrons are removed • general increase because the same number of protons are holding fewer and fewer electrons • general increase because electrons are being removed from an increasingly positive ion neutral answers – less shielding / electrons are closer to the nucleus	1			1		
			II	award (1) for either of following • sharp rises due to electron being removed from an inner shell / a shell which is closer to nucleus • sharp rises due to electron being removed from a shell which is closer to nucleus so has less shielding do not accept reference to sub-shells	1			1		